A Unified Science of Perceptual/Motor Control through Sensor-Grounded Ontology

FreeMoCap Foundation, Inc.

Written Proposal to the NSF X-Labs Initiative

> The mission of the proposed NSF X-Lab is to build a new class of scientific instrument — the complete, calibrated, synchronized empirical capture of agent-environment interaction — and the open organization that can sustain it, unifying the study of perception and action across humans, animals, and machines into a single science of real-world sensorimotor control.

Every living and engineered agent solves the same problem: it senses a thin slice of the energy available in its environment through a limited set of imperfect transducers, and on that basis generate reaction forces against an available substrate to propel its toward its goal. Information flows in, forces flow out; the brain exists to yank the bones around.

Perceptuomotor neuroscience, musculoskeletal biomechanics, and legged robotics have each built vibrant research programs and technically marvelous instruments — Neuropixels, markerless motion capture, force-plate gait labs, motor-unit EMG [TODO - Improve this list, focus on measurement rather than company/tool (kinetics, kinematics, binocular gaze, neurons (ephys neuropixel and w/e a 1Photon miniscope does), muscle activity etc, like in `fracture_need_b`)]— around individual threads of the perception-action loop. Each field measures its thread with extraordinary precision.

But no instrument captures the loop whole. The threads are measured in incompatible coordinate frames, on unsynchronized clocks, under semantic schemes that do not talk to each other, so the single most basic question — what did this agent see, and what did it therefore do — has no calibrated, unified answer. The fracture is in the measurement itself, not just in the sociology of the fields. *[TODO - this sucks, improve it. ok content, needs rewrite]*

And the instrument that would close it falls through every existing crack: too integrative for a single-modality grant, too infrastructural to be a paper, too long-horizon for a graduate degree, too open to be a product. The academic incentive structure rewards the thread, not the loom. *[TODO - This also sucks. content ok, needs rewrite]*

What is needed is a new class of scientific instrument: a calibrated, synchronized, semantically-unified recording of the complete agent-environment interaction — every empirically available channel of an agent's sensory/perceptual input and motor/mechanical output captured, calibrated, and processed with metrologically-grounded, truth-preserving pipelines into a unified scientific ontology *[TODO - Clarity, Awk, Run-on, overloaded]*.

We call the instrument a **Dense Empirical Capture Volume**: a densely instrumented region of real-world space engineered to record every measurable channel of the agent-environment interaction at once — binocular gaze and reconstructed retinal input, full-body kinematics and kinetics, muscle and motor-unit activation, and central and peripheral neural activity — all spatially calibrated, temporally synchronized, and expressed in a single sensor-grounded ontology so the channels are directly commensurable rather than merely co-recorded.

Because the instrument is defined by its ontology rather than its hardware, measurements for differnt sensor systems may hydrate the same model. Camera- and IMU-based motion capture each measure the same `Human` kinematics with very different uncertainty profiles, so sensor-fused estimate will be far more reliable than then either sensor's estmate alone, but all three estimtes (camera, imu, or hybrid) define the exact same `Human` model. The shared ontology aligns across different combinations of sensors, and across different generations of the same transducer. A `Head` has `Eyes` even if the participant did not wear an eye-tracker. An `Eye` defines `Torsion`, and `LensAccomodation`, even if the current era of eye trackers can only estimate `Adduction` and `Elevation.` And then once we build a better eye tracker that **can** record torsion [TODO - Reference `Outcomes` section where we talk about eye tracker plans], we can train a model relating `Torsion Adduction Elevation` and use it to backfill older data as a way to test next generation of theories against the previous generation's data. *[TODO - Good content, some of the detail may want to move to a different section.]*

The FreeMoCap X-Lab will build a flagship facility for a large-scale DECV [DECV-L for Large] facility in **Boston, MA**. The DECV-L will be a warehouse-scale modular instrument designed for recording full-body kinematics, kinetics, and binocular eye tracking of freely moving humans within configurable open volume (Building on the PI's indoor AR/MoCap history[1–4]).. Critically, the DECV-L will be large enough to fully cover smaller DECVs representative of the kinds that would be build by other research labs and FreeMoCap users, so the DECV-L can serve both as a flagship representative of the DECV instrument, but also a validation and meta-experimental platform to assist in the dissemination of this platform to downstream consumers. In addition the DECV-L will large enough to assist in the development and validation of a wearable IMU-based mocap suit plus binocular eye tracker and head-mounted world camera (DECV-W for Wearable), buiding on the lineage of the PI's outdoor gaze/gait control[5] and retinal optic flow research[6–9] - something something mention the drone swarm thing or leave to later?)

Around the human flagship sits a federated network of partner labs running **functionally-equivalent instruments** on their own model systems — the invasive neural modalities that cannot go on humans (Neuropixels, miniscope calcium imaging) living on the animal branches, plus prosthetic and robotic platforms as engineered agents in the same ontology. These are not hypothetical: we have already built plausibly best-in-world eye-trackers for ferret and mouse — a ferret rig fusing three-camera gaze, full-body kinematics, AR display, and world cameras in one calibrated system is, in effect, a miniature animal-scale instance of the whole flagship, already demonstrated and now integrating electrophysiology.

The flagship wings and the partner labs are a single distributed instrument, not a consortium, because they share one sensor-grounded ontology, one metrological traceability chain (every site's measurements uncertainty-tagged and commensurable across species and equipment tiers), one comparative design, and one annual convening that keeps the federation aligned.

The FreeMoCap Foundation has already demonstrated the model at smaller scale: an open-source markerless motion-capture instrument, built and maintained outside academia, now used by thousands of researchers who could not otherwise access the capability. The X-Lab is that proven approach — open, tool-first, community-grounded — applied to the far larger target of the complete capture volume, at a scale that only dedicated, autonomous funding can reach.

An instrument like this cannot be built inside the institutions that need it. Sustained tool-building does not amortize over the paper-and- grant cycle: it needs full-time engineers held for years, not students rotating out on degree timelines, and it needs a home whose success metric is a working, shared instrument rather than publication count. The FreeMoCap Foundation was founded in 2021 precisely to occupy this gap — a 501(c)(3) built to develop open scientific instruments as its primary output.

The end-state is a single empirical science of sensorimotor control. Today, perceptuomotor neuroscience, musculoskeletal biomechanics, mobile robotics, and agentic AI study the same loop with incompatible tools and vocabularies. A shared instrument makes their measurements commensurable — a finding in one becomes evidence in another — collapsing four parallel literatures into one cumulative field studying real-world behavior in real and naturalistic environments.

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Figure 1: Blah blah blah captions. Blah blah blah captions. Blah blah blah captions. Blah blah blah captions. Blah blah blah captions.

1. Technology Landscape

- Existing tools and methods
  - Myriad single-modality hyperspecialized research threads
    - OpenSim - Widely used but universally hated by practictioners
    - DeepLabCut - Global success on paper, universally a headache in practice
    - etc
  - MuJoCo/IsaacLab Driving robotics
    - Integrate NeuroBehaioral Data into those sinulation/RL training enviornments to cross-align perceptuomotor neuroscience, HNHA biomechanics, robotics control research (e.g. Cassie from Guinea Fowl)
- New approaches to complexity management
  - Ontology/Entity Management (Anduril/Palantir/Maven Smart System)
  - Industry and Military technology applied to scientific domain
  - Want to establish an operational ontology as backbone of this tool, following powerful patterns established by (Anduril/Palantir/Maven) systems
- FreeMoCap and its landscape
  - Established Tool with Growing International Community
  - Existence is proof-of-concept of approach, XLabs scale support would accelerate FreeMoCap model into the global scale
- Laser Skeleton research
  - [5–7]
  - etc
  - Ferret work
    -

2. Outcomes

- Cool Tool - ~~Skinner~~SkellyBox: Multiscale modular empirical capture arena
  - Interoperabilty across species, domains, and timescales
    - A member of a mouse lab can walk into a marmoset lab and expect their pipelines will still work
    - Perceptuomotor behavior and kinematic/kinetic analysis directly interoperable with robotics models (via MuJoCo and Nvidia Isaac Gym) - Data for reinforcement learning models, Controls to interpret HNHA behavioral data
    - Data models alignment across timescales - developmental/longitudinal (months/years/lifespan) timescales aligned with perception/action timescales (seconds/minutes)
  - Composable complexity
    - Careful ontology allows for complexity via comopsible data sources and assocaited downstream entities
    - Adding new sensor modalities into established scene, framework handles spatiotemporal alignment/calibration, unit conversions, resampling, etc
- Organize and operationalize a new culture of science around discussion/management/extension shared platform tool
  - In the form of an annual Congress meeting
  - Organizing the culture of science around the discussion and advancement of shared ontologies rather than an endless stream of 8-10 page pdfs
  - Break the Skill Ceiling in academic science
    - Build tool to allow multi-decade continuous education across disciplines and experience (same tool in high school as used in federally funded research lab)
    - Shared toolset encourages natural blending of disciplines while encouraging and reward higher specialized exploration (esp when that research produces actionable results across different dimensional domains)
- Mesoscale revolution
  - Allow advances in reductionist microscale research to align with human-scale interests
  - Align Perceptuomotor Neuroscience with humanoid robotics, allow transfer, technology, and technique transfer between those fields

2.1. Timeline

Phase	Name	Description
1	**Organize**	Set up Organization (Phase 0) — Solidify release of FreeMoCap v2
2	**Architect**	Develop core software architecture, define key ontologies and entity/action pairings via use-oriented development (ferret, mouse, guinea fowl, marmoset, human research). Design and prototype instrumentation (mocap/eye cameras, electrode arrays and miniscopes, motor-unit EMG, environmental control systems)
3	**Execute**	Integrate new architecture into global/public facing community of users (presented as FreeMoCap v3). Iterative design of instrumented system
4	**Stabilize**	Establish OODA loop within expected standard operations
5	**Expand**	Once stabilized, expand key performance metrics
6	**Align**	Focus on building deeply integrated community of users and contributors (Transition BDFL ⟶ BDFaSPoT)
7	**Establish**	Focus on long term stability and healthy growth
8	**Maintain**	Beyond proposal timeline — Establish organization practice in service of unbounded infinite timeline. Apply winning model to support extending 'mesoscale research revolution' to all biological sciences (e.g. integrating deeper microbio/biochem insights beyond the scope of the present perceptuomotor/biomech/robotics focus)

Table 1: Timeline of Phased Milestones

3. Senior/Key Personnel Qualifications

3.1. Jonathan Matthis, PhD [Principle Investigator]

3.2. EI

3.3. AC,PhD

3.4. RR, CPA or w/e

Name / Role	Qualifications
Jonathan Matthis, PhD President/CEO	• Founder and chief maintainer of FreeMoCap project • Left tenure track academic research position precisely because the institution could not support the work that needed to be done • Decades experience integrating visual neuroscience with biomechanics/ robotics[5–7]
EI CTO	• Decades of experience in software Industry • Chief architect and CTO for established companies • Represents expertise truly absent from academic contexts • Ensures world-class enterprise-level software capability • Will build training/workshop/hackathon backend to inject EI-level software development expertise into broad community of science
AC, PhD CSO	• Expertise in validation of clinical research tools and bench-to-bedside development • [CITE DISSERTATION] • Experience in neuroprosthetics design
RR CFO	• Bookkeeping • NSF Grant management • Finances • Entrepreneurship

Table 2: Senior/Key Personnel Qualifications

4. Team Capabilities Statement

4.1. Senior Team Capabilities

- Jon - done it before in humans, doing it in ferrets and mice
- AC - focusing clinical adoption and bench-to-bedside transitions
- EI - Deep professional experience in Tech Sector, high level software architecture experienced necessary to develop truly global scale enterprise level of software in open-source/academic contexts
- RR - Organizational health and entrepreneurial development

4.2. Collaborators

Initials	Model / Domain	Expertise
BS	Ferrets	Visual/Perceptual neuroscience, ephys
MD	Guinea Fowl	Musculoskeletal Biomechanics, muscle units
DF	Mice	System Biology
AH	Marmosets / NHP	Ephys
MH	Human	Vision and CPS
Mayo Clinic	Human	Brain scans and implanted sensors
BD, GN/RAI, CH, JH	Robots / Fabrication	Control theory
AA	Physical facilities	Rapid iteration, recruiting

Table 3: Collaborator Network

4.3. Governance

- Most successful FOSS project operate under a Benevolent Dictator For Life (BDFL) Governance (e.g. Linus Torvald at Linux)
- This is best when an organization is young, but later on centralization is a concern control should melt into the community of users (see Ton Roosenthaal's step down from Blender lead position)
- We will use Benevolent Dictator for a Set Period of Time (BDfaSPoT)
  - Transition from centralized control: President -> President + Board -> to President + Board (C-Suite) + Community DAO
    - President maintains at least 51% control until year 5, then optional transition to 50/50 President/Board+Community
- Plan changes via public Requests for Comments (similar to Python PEP system)
  - Provide public space for feedback on planned features, allows transparency without creating a bottleneck
- In 2nd year, create Developer Fund and Community Grants program
  - Stakeholders vote on proposals and features, via DAO like voting
- In 4-5th year, extend this system to handle community steering of org

4.4. Autonomy

- We are already autonomous, as of 2 weeks prior to the deadline of this proposal.
  - This project will happen no matter what, but with XLabs funding, we could truly change the world

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